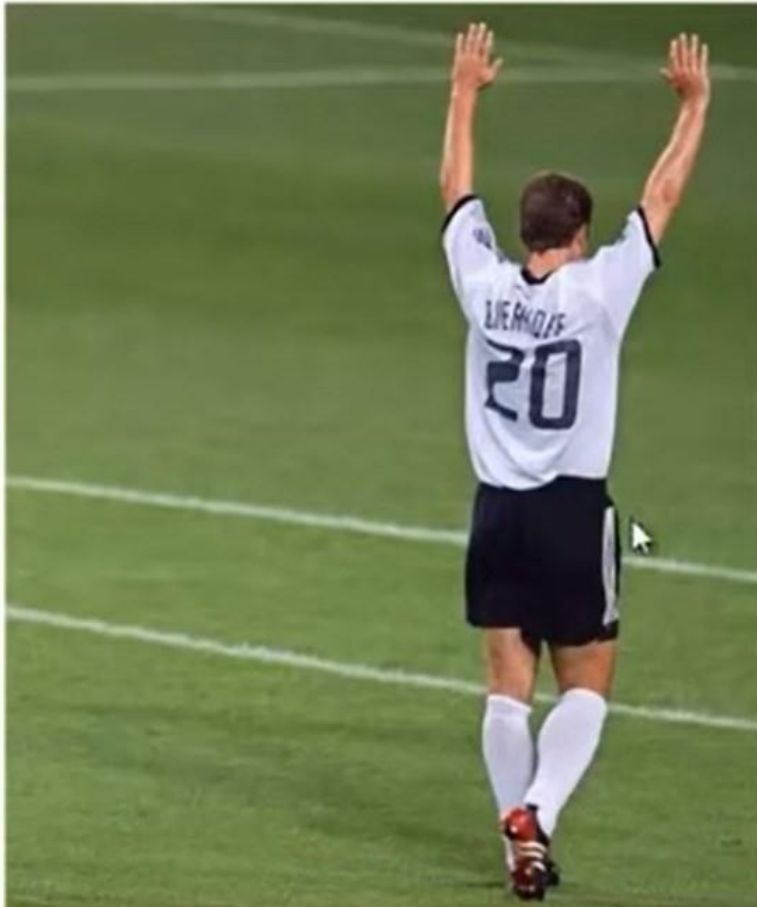




CAMERA CALLIBRATION

Application – Object Transfer

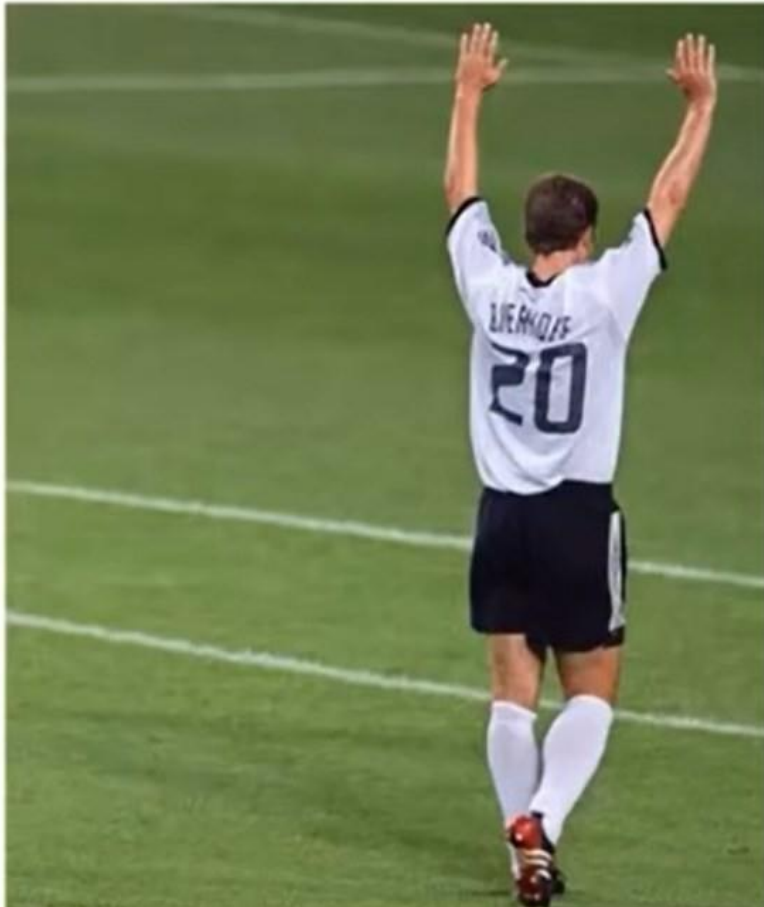


Source Image



Target Image

Application – Object Transfer

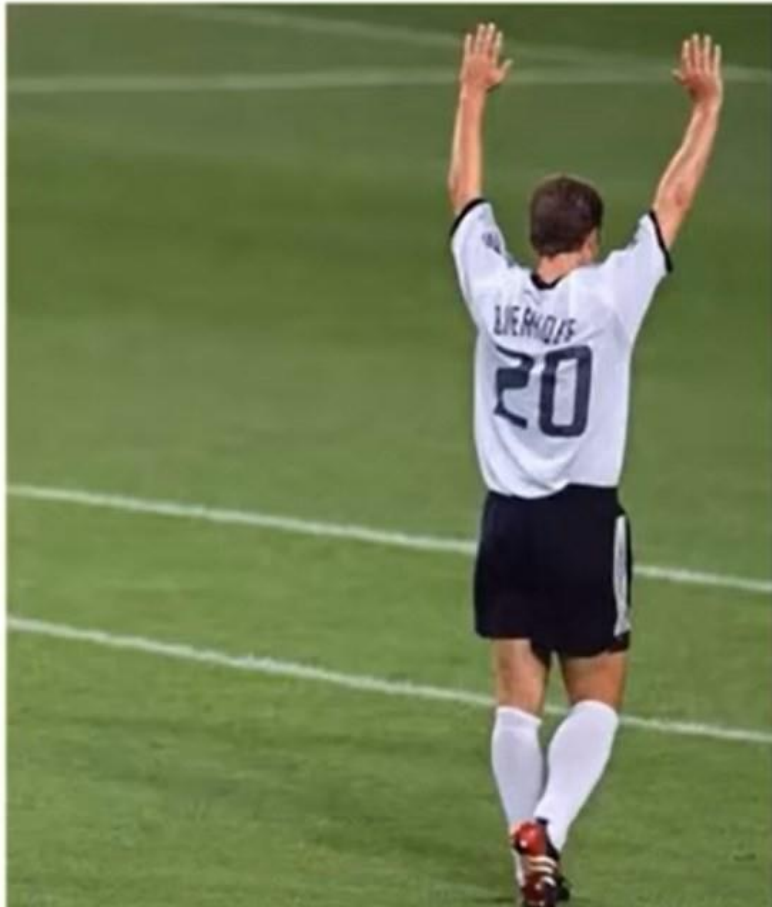


Source Image



Target Image

Application – Object Transfer



Source Image



Target Image

Camera Calibration – Overview

A thick yellow horizontal bar spans the width of the slide, with a vertical yellow bar extending downwards from its right end.

- Camera calibration is the process of determining the intrinsic and extrinsic parameters of a camera.
- These are essential for accurate mapping between the 3D world and 2D image captured by the camera.
- The process involves calibrating the camera's internal parameters (intrinsic) and its position in the world (extrinsic).

Intrinsic Parameters

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- Intrinsic parameters are the internal characteristics of a camera that remain constant regardless of its position or orientation in 3D space.
- These parameters are specific to each camera and are essential for accurately projecting 3D points onto the 2D image plane.

Intrinsic Parameters

- The most important intrinsic parameters include:
 - **Focal Length (f):** The focal length is the distance between the camera's optical center and the image sensor (or film plane). It determines the magnification and field of view of the camera.
 - **Principal Point (c_x, c_y):** The principal point is the coordinates of the image center in the image plane. It represents the location of the optical axis's intersection with the image sensor.
 - **Lens Distortion:** Real-world camera lenses often introduce optical distortions, such as radial distortion (barrel or pincushion distortion) and tangential distortion. These distortions cause a deviation from the ideal pinhole camera model, affecting the accuracy of measurements in the image.

Extrinsic Parameters

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- Extrinsic parameters describe the camera's position and orientation in 3D space relative to a known coordinate system.
- Unlike intrinsic parameters, extrinsic parameters can change as the camera moves or rotates.
- They are crucial for understanding the camera's viewpoint with respect to the 3D scene being captured.

Extrinsic Parameters

- The main extrinsic parameters are:
 - **Translation (T_x , T_y , T_z):** The translation parameters represent the position of the camera's optical center relative to the origin of the world coordinate system. They specify how far the camera is displaced in the x, y, and z directions.
 - **Rotation (θ_x , θ_y , θ_z):** The rotation parameters define the camera's orientation in 3D space. They represent rotations around the x, y, and z axes, which determine the camera's tilt, pan, and roll, respectively.

Camera Models

- Camera models represent the mathematical equations that describe how a 3D point in the world is projected onto the 2D image plane of the camera.
- The most common camera model is the pinhole camera model, which assumes that light rays from the 3D scene pass through a single point (the camera center) before forming an image on the image sensor.

Purpose

A thick yellow horizontal bar spans the width of the slide, with a vertical yellow bar extending downwards from its right end.

- Calibration is necessary because real-world cameras often introduce distortions due to lens imperfections and optical aberrations.
- Calibration corrects optical distortions, enabling precise measurements from images and accurately measure objects in the scene.
- It enables various computer vision tasks such as 3D reconstruction, camera pose estimation, image rectification, augmented reality, and camera tracking.

Applications

- Camera calibration is necessary for various computer vision tasks, such as:
 - **3D Reconstruction:** Accurately determine the 3D coordinates of points in the scene.
 - **Camera Pose Estimation:** Find the position and orientation of the camera with respect to the scene.
 - **Image Rectification:** Correct image distortions for accurate measurements and matching.
 - **Augmented Reality:** Precisely overlay virtual objects onto real-world images.
 - **Camera Tracking:** Track camera movement for applications like visual odometry.

Geometric Transformations

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- Geometric transformations are used in computer graphics and computer vision to map points from one coordinate system to another.
- It preserves straight lines and parallelism, but it does not necessarily preserve angles, lengths, or areas.
- They are widely used because they are computationally efficient and can represent various common geometric operations.

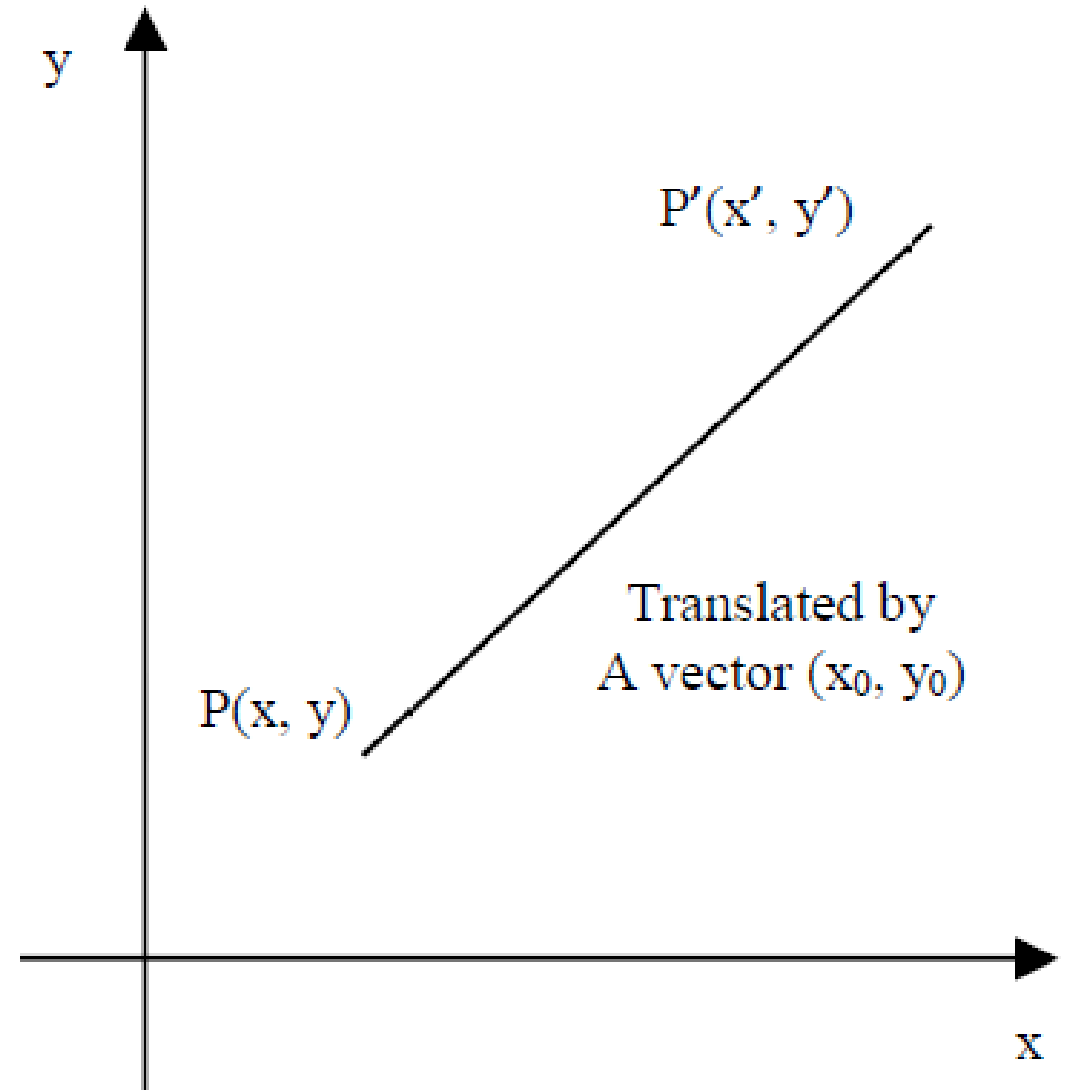
2D Transformations

- Translation

- $x' = x + x_0$
- $y' = y + y_0$

- It can be written as column vector:

$$P = \begin{bmatrix} x \\ y \end{bmatrix}, P' = \begin{bmatrix} x' \\ y' \end{bmatrix}, T = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$
$$P' = P + T$$



2D Translation

Then the translation operation can be written as:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

In homogeneous coordinates:

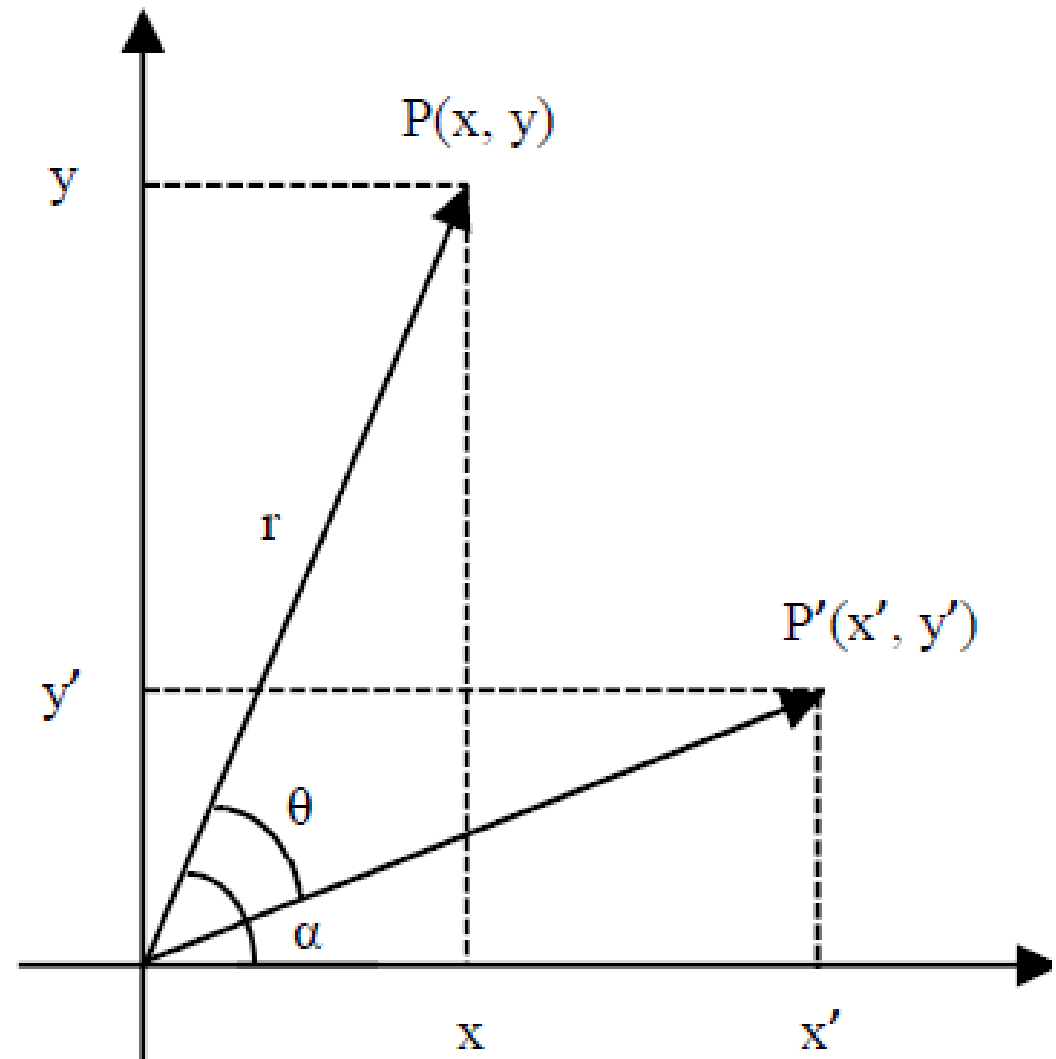
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

 Symmetric expression

2D Rotation

- Now, let us consider the concept of rotation of a point $(x; y)$ by an angle in the clockwise direction

- $x = r \cos \alpha$
- $y = r \sin \alpha$



2D Rotation

$$x' = r \cos(\alpha - \theta) = x \cos \theta + y \sin \theta$$

$$y' = r \sin(\alpha - \theta) = y \cos \theta - x \sin \theta$$

In matrix form,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Scaling

- Next, we consider the case of scaling of a point by a factor S_x and S_y along the direction x and y , respectively.
- In matrix form, a scaling operation can be represented as:

$$\begin{bmatrix} x \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

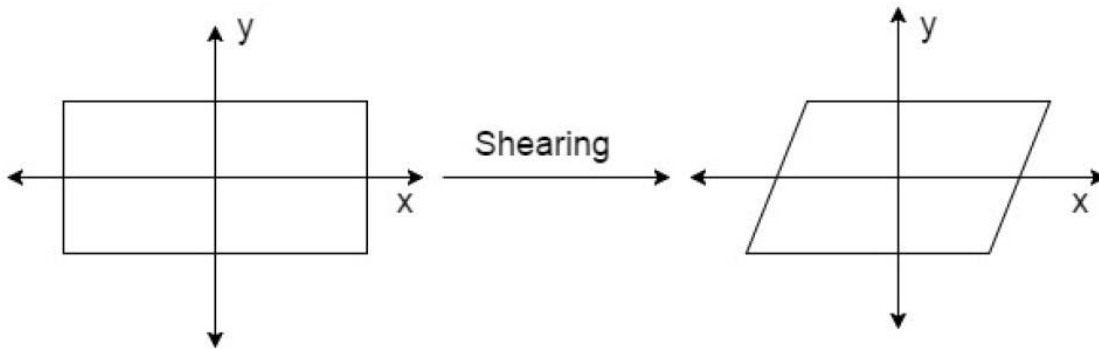
2D Shearing

- If the scaling is not uniform for the whole object, then it is called shearing.
- For example, the shearing parallel to the x-axis can be represented as:

$$x' = x + ky \text{ and } y' = y$$

$$\begin{bmatrix} x' \\ y \end{bmatrix} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Shearing



$$\begin{bmatrix} 1 & h_x \\ h_y & 1 \end{bmatrix}$$

In homogeneous coordinate system, this transformation can be written as:

$$\begin{bmatrix} 1 & h_x & 0 \\ h_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3D Transformations

- Now, we can extend all of the above-mentioned ideas to 3D in the following way. Matrices can be used to represent 3D affine transforms in homogeneous form. All the 3D points are converted to homogeneous coordinates as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Translation

Translation of a point is given
by:

$$x^* = x + x_0$$

$$y^* = y + y_0$$

$$z^* = z + z_0$$

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & x_0 \\ 0 & 1 & 0 & y_0 \\ 0 & 0 & 1 & z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Translation

- The transformation matrix for translation is now converted into a square matrix. So, the transformation matrix for translation is given by:

$$T = \begin{bmatrix} 1 & 0 & 0 & x_0 \\ 0 & 1 & 0 & y_0 \\ 0 & 0 & 1 & z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Scaling

- The transformation matrix for scaling is:

$$S = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where, S_x : Scaling in x -direction
 S_y : Scaling in y -direction
 S_z : Scaling in z -direction

3D Rotation

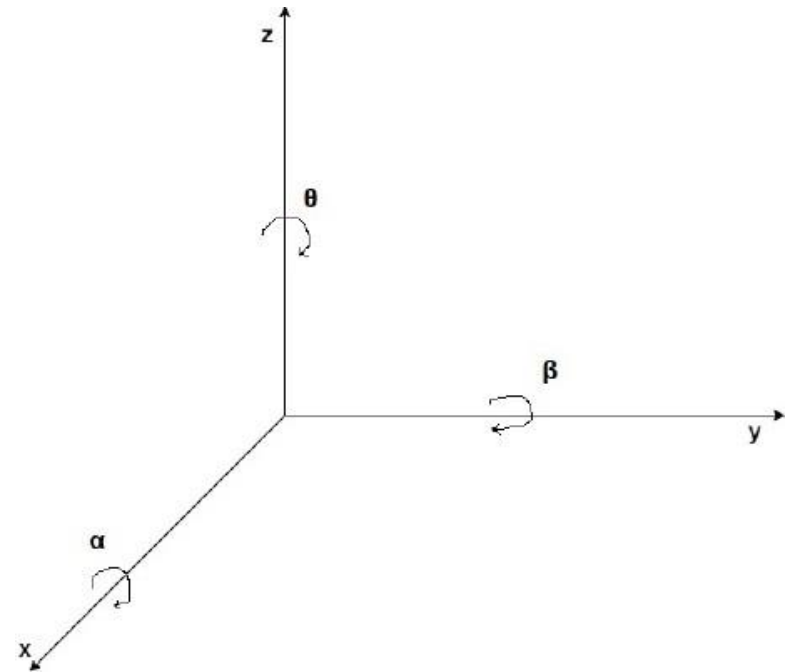
- Let us consider three cases of rotation, i.e., rotation around z-axis, rotation around x-axis and rotation around y-axis.
- For rotation around z-axis, only x and y coordinates change, and z coordinate remains the same.
- Rotation around z-axis is the rotation in a plane which is parallel to the x-y plane

3D Rotation

$$R_\theta = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$R_\alpha = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$R_\beta = \begin{bmatrix} \cos\beta & 0 & \sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$



3D Shear

Finally, the transformation matrix for shear is given by:

$$\begin{bmatrix} 1 & h_{xy} & h_{xz} & 0 \\ h_{yx} & 1 & h_{yz} & 0 \\ h_{zx} & h_{zy} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Transformation

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- The perspective projection of a camera can be defined in terms of frame of reference of a camera. It is important to find location and orientation (extrinsic parameters) of the reference frame of the camera with respect to a known world reference frame.
- The parameters which describe the transformation between the reference frame of a camera and world coordinates are:

3D Transformation

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- 3D translation vector T , which is the relative displacement of the origins of the reference frame of a camera with respect to a known world reference coordinate.
- 3 x 3 rotation matrix R which is required to align the axes of the two frames (camera and world) onto each other.

Calibration

